

Urban Traffic Congestion Analyzer + Signal Timing Recommendation (Offline)

1. Problem Overview

The PDF asks for an offline analytics tool that identifies congestion peaks and bottlenecks from historical traffic flow data and recommends signal timing adjustments or coordination suggestions. It is decision support only: no real-time control, no guaranteed improvement claims, and all simulations must clearly state assumptions. In practical terms, the task is to help traffic engineers decide how to retune signals before any field deployment.[jetirAI-Domain-X-Challenge-2026-Problem-Statements.pdf](#)

2. What the Problem Is Asking

The problem is asking you to analyze how traffic congestion changes by time of day and road segment, then use historical counts to suggest better signal timing. The expected output is not just a chart; it should identify where queues and delays are concentrated and explain why certain green split changes or coordination ideas might help. The solution should also include a before-vs-after style estimate using a simple simulation or queueing model.[AI-Domain-X-Challenge-2026-Problem-Statements.pdf](#)

3. What Is Expected

The PDF expects four outputs: peak congestion patterns, bottleneck identification, timing recommendations with rationale, and dashboard visualization. It also expects a simple simulation showing likely improvement after the proposed changes. The recommendations should be understandable to traffic engineers and should include explicit assumptions.[AI-Domain-X-Challenge-2026-Problem-Statements.pdf](#)

4. What Is Not Expected

This is not a live traffic signal control system. It should not promise guaranteed improvement because actual results depend on field conditions, detector quality,

pedestrians, incidents, and implementation constraints. It should also avoid unsafe or overly confident claims.[AI-Domain-X-Challenge-2026-Problem-Statements.pdf](#)

5. Business Context

Cities want to reduce delay, improve travel time reliability, and make better use of existing infrastructure. Static signal timings often become inefficient as traffic demand changes by hour or day. This makes offline analytics valuable because it can identify where current timings are mismatched with demand.[jetirAI-Domain-X-Challenge-2026-Problem-Statements.pdf](#)

6. Engineering Context

From an engineering perspective, this is a signal timing and coordination problem, not just a data visualization problem. The key goal is to allocate limited green time across competing movements in a way that reduces delay and queue growth. That means understanding saturation flow, cycle length, split, offset, and progression across intersections.[jetir](#)

7. Core Traffic Concepts

Congestion

Congestion happens when traffic demand exceeds the ability of the road or intersection to serve vehicles. It matters because delay, queueing, and unreliability rise quickly once demand approaches capacity. Example: an intersection approach with insufficient green time will accumulate a queue every cycle.[jetir](#)

Bottleneck

A bottleneck is the part of the network where flow is restricted more than elsewhere. It matters because the whole corridor can be affected by one constrained movement. Example: a short left-turn bay may block a through lane.[jetir](#)

Queue Formation

Queues form when arrivals exceed discharge for a period of time. It matters because queue growth is the clearest sign of inadequate capacity or poor timing. Illustration: if 20 vehicles arrive per minute but only 15 can be served, the queue grows by 5 per minute.[jetir](#)

Delay

Delay is the extra time spent compared with free-flow or ideal movement. It matters because delay is the main performance indicator for intersections. Example: vehicles waiting at a red light experience control delay.[jetir](#)

Throughput

Throughput is the number of vehicles actually served in a given period. It matters because a good signal plan should increase served traffic or reduce delay without causing spillback.

Illustration: more effective green allocation can raise throughput on a congested movement.[jetir](#)

Capacity

Capacity is the maximum flow that can pass under prevailing conditions. It matters because congestion begins when demand exceeds capacity. Example: an approach has a finite discharge rate even if demand is much higher.[jetir](#)

8. Signal Timing Concepts

Signalized Intersection

A signalized intersection uses traffic lights to manage conflicting movements. It matters because signal timing determines how available capacity is shared. Example: one phase may serve north-south through traffic while another serves east-west turns.[jetir](#)

Phase

A phase is the set of movements that receive green together. It matters because phases divide the cycle into service windows. Example: protected left turns may need their own phase.[jetir](#)

Green Time

Green time is the period during which a movement is allowed to proceed. It matters because more green generally means more vehicles can be served. Example: increasing green for a heavy approach may reduce its queue.[jetir](#)

Red Time

Red time is the period during which a movement must stop. It matters because it directly contributes to delay and queue build-up. Example: a minor approach may wait through a long main-road green.[jetir](#)

Cycle Length

Cycle length is the total time for one complete sequence of phases. It matters because it sets the overall rhythm of the intersection and affects delay and coordination. Example: a 90-second cycle serves all phases once every 90 seconds.[jetir](#)

Offset

Offset is the time difference between the start of green at successive intersections. It matters because good offsets help vehicle platoons progress along a corridor with fewer stops.

Example: if vehicles reach the next signal when it turns green, progression improves.[jetir](#)

Coordination

Coordination synchronizes multiple intersections so traffic can move more smoothly along a corridor. It matters because one intersection cannot be optimized in isolation if nearby signals are coupled. Example: coordinated signals can create a green band for a dominant travel direction.[jetir](#)

Green Wave

A green wave is a coordinated timing pattern that allows vehicles to pass through several signals with minimal stopping. It matters because it reduces stops and travel time on arterials. Example: a commuter corridor timed for peak-direction progression.[jetir](#)

9. Static vs Adaptive Signals

Static signals use fixed timings or time-of-day plans that only change periodically. Adaptive signals adjust timings based on real-time detector data and changing demand. This hackathon problem is closer to offline planning than live adaptive

control.[AI-Domain-X-Challenge-2026-Problem-Statements.pdfjetir](#)

10. Offline vs Online Optimization

Offline optimization means analyzing historical data and recommending a timing plan for later use. Online optimization means the controller updates timings continuously as traffic changes, which is outside the problem scope. Offline is simpler, more transparent, and better suited to a hackathon MVP.[AI-Domain-X-Challenge-2026-Problem-Statements.pdf](#)

11. Current Industry Workflow

Traffic engineers usually start with counts, turning movements, road geometry, and existing timing plans. They identify critical movements, then compute or test candidate signal plans using engineering methods such as Webster-style timing and coordination analysis. Final changes are normally reviewed by humans because signal timing affects safety, operations, and policy decisions.[jetir](#)

12. Common Limitations

Traffic demand changes by hour, day, season, incidents, and land use, so a fixed plan can become outdated. Detector coverage is often incomplete, and many intersections rely on sparse or noisy data. Coordination also creates tradeoffs, because helping one movement can hurt another.[jetir](#)

13. Data Requirements

The PDF explicitly mentions traffic counts by intersection, vehicles per minute, time of day, road metadata such as lanes and speed limit, turning ratios, and optional incident events like roadwork or accidents. A useful implementation would also benefit from intersection IDs, approach labels, phase IDs, signal plans, queue length, and travel time if available. Synthetic data is acceptable under the problem's guidance.[AI-Domain-X-Challenge-2026-Problem-Statements.pdfjetir](#)

14. Traffic Engineering Basics

Saturation Flow

Saturation flow is the maximum discharge rate during a sustained green. It matters because it is the basis for calculating how much traffic a phase can serve. Example: if a lane discharges 1,800 veh/h under green, that is its saturation flow.[jetir](#)

Volume-to-Capacity Ratio

The volume-to-capacity ratio compares demand to service capacity. It matters because a ratio near or above 1 indicates likely queue growth. Example: 900 veh/h demand over 1,800 veh/h capacity gives a ratio of 0.5.[jetir](#)

Webster Method

Webster's method is a classic fixed-time signal design approach that estimates a near-optimal cycle length and allocates green time by demand. It matters because it is a standard offline baseline for timing design. A common form is

$$C_0 = (1.5L + 5) / (1 - Y)$$

C

0

$$= (1.5L + 5) / (1 - Y), \text{ where}$$

L

L is lost time and

Y

Y is the sum of critical flow ratios.[jetir](#)

Worked Example

If total lost time is 16 seconds and the sum of critical flow ratios is 0.80, the estimated Webster cycle length is about 145 seconds. That cycle would then be split among phases according to their critical demand share after accounting for lost time. This is a practical baseline, not a guarantee of optimal performance.[jetir](#)

15. Stakeholder Analysis

Traffic engineers want lower delay, better progression, and clearer evidence for timing changes. Municipal corporations want public mobility improvements and efficient use of infrastructure. Smart city authorities want corridor-level analytics and citywide visibility, while traffic police care about incident response and field practicality.[AI-Domain-X-Challenge-2026-Problem-Statements.pdfjetir](#)

Emergency services care about reliable access and fast movement during urgent situations. Citizens care about shorter commute times, fewer stops, and more predictable travel. Because these goals can conflict, signal timing is both an engineering and a policy issue.[AI-Domain-X-Challenge-2026-Problem-Statements.pdfjetir](#)

16. Existing Systems

SCATS is a real-world adaptive signal system used in many cities; its official site reports deployment at more than 60,000 intersections across 200 cities and 30 countries. FHWA's

traffic signal timing manual explains the engineering basis of coordination, including cycle length, splits, offsets, and progression. The key takeaway is that the hackathon problem sits in the offline planning layer, not the live control

layer.[AI-Domain-X-Challenge-2026-Problem-Statements.pdf](#)[jetir](#)

17. Practical Opportunities

The best opportunity is to explain which movements are critical and why they dominate the timing decision. Another opportunity is corridor-level reasoning, because optimizing one intersection alone can hurt downstream progression. A clear before-vs-after simulation with explicit assumptions can make the solution more

credible.[AI-Domain-X-Challenge-2026-Problem-Statements.pdf](#)[jetir](#)

18. Common Mistakes

One common mistake is assuming more green automatically solves congestion. In reality, bad allocation can shift queues or worsen coordination. Another mistake is ignoring offsets and optimizing intersections independently. A third mistake is presenting simulation output as guaranteed improvement, which the PDF does not

allow.[AI-Domain-X-Challenge-2026-Problem-Statements.pdf](#)[jetir](#)

19. Key Takeaway

This problem is an offline traffic-signal analytics task built around historical counts, simple queueing or simulation logic, and explainable timing recommendations. It is not a real-time controller and not just a dashboard task. The strongest solutions will be transparent, corridor-aware, and careful about

assumptions.[AI-Domain-X-Challenge-2026-Problem-Statements.pdf](#)[jetir](#)

20. Source Notes

Primary source: AI Domain-X Challenge 2026 PDF, Problem Statement #18. External reference: FHWA Traffic Signal Timing Manual, Chapter 6 on coordination. External reference: official SCATS homepage.[AI-Domain-X-Challenge-2026-Problem-Statements.pdf](#)[jetir](#)



2 sources

